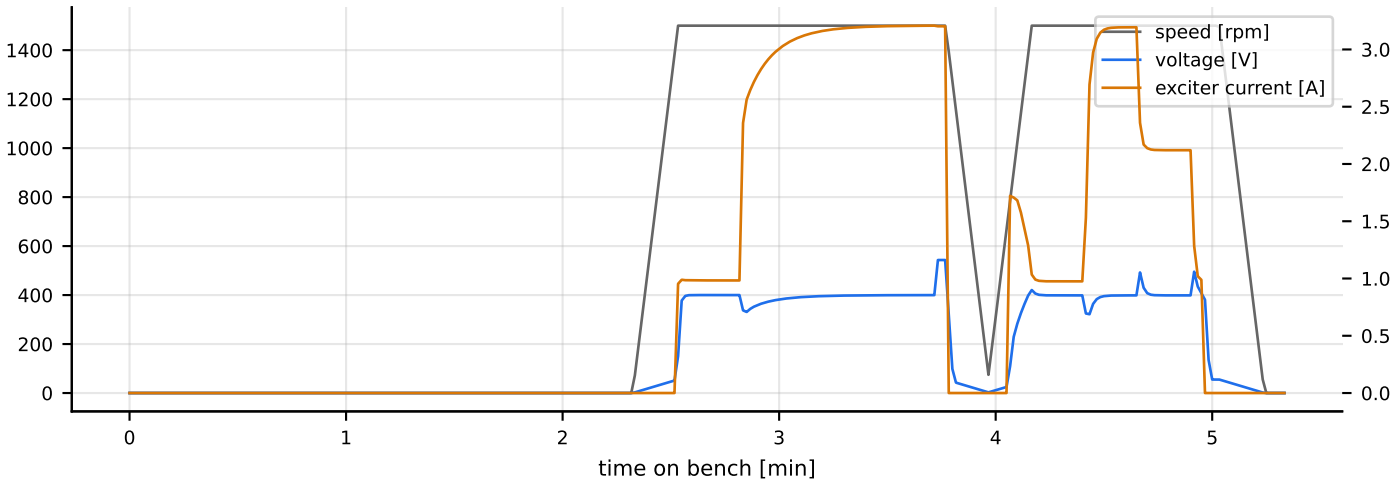
Routine test report

Low-voltage alternator, production test bench (simulation)

Serial	EX-0002-FAIL
Frame	medium
Test connection	400 V star, 50 Hz
Test load	640 kW
Test type	routine test
Duration on bench	5.3 min

FAIL

Check	Value	Unit	Limits	Result
Stator winding resistance at 20 degC	0.00415	ohm	0.00399 .. 0.00441	OK
Main rotor resistance at 20 degC	1.868	ohm	1.805 .. 1.995	OK
Exciter stator resistance at 20 degC	9.484	ohm	9.025 .. 9.975	OK
Exciter rotor resistance at 20 degC	0.211	ohm	0.199 .. 0.221	OK
Insulation resistance before HV	556.29	Mohm	>= 100.00	OK
Dielectric withstand, peak leakage	21.22	mA	<= 50.00	OK
Phase sequence	correct			OK
Voltage unbalance, no load	0.07962	%	<= 0.500	OK
Exciter field current, no load at Un	0.983	A	0.675 .. 0.825	**FAIL**
AVR supply voltage at no load	150.21	V	132.00 .. 168.00	OK
Exciter field current on load at Un	3.203	A	2.145 .. 2.621	**FAIL**
Load cut-off voltage / Un	1.358	-	1.250 .. 1.400	OK
Voltage build-up frequency with AVR	28.72	Hz	<= 40.00	OK
No-load voltage with AVR, deviation	-0.259	%	-1.000 .. 1.000	OK
Voltage regulation at PF 0.8	-0.05762	%	-1.500 .. 1.500	OK
Voltage regulation at PF 1.0	-0.05551	%	-1.500 .. 1.500	OK
CT secondary current and polarity	1.001	A	0.900 .. 1.200	OK
Remanent voltage / Un	13.78	%	>= 10.00	OK
PT100 resistance	108.81	ohm	105.00 .. 115.00	OK
PTC resistance	64.16	ohm	20.00 .. 250.00	OK
Space heater resistance	211.48	ohm	196.79 .. 226.41	OK



Non-conformity: Excitation needs too much field current: check rotating diodes and exciter -> REPAIR

Simulated unit and bench. Generic limits, not a manufacturer's values.